Fundamentals of Quantum Mechanics

Study Material Developed from PPT
This document is developed from the uploaded PPT (Quantum Science and Technology, Unit 1) as an expanded set of lecture notes. It is intended for B.Tech/M.Tech students.

Course Outcome (CO1): Explain core principles of quantum mechanics and their technological implications.

Learning Outcomes

- LO1: Explain the fundamental concepts and historical development of quantum mechanics.
- LO2: Apply the Schrödinger equation to solve simple quantum mechanical systems.
- LO3: Analyze quantum states using operators, observables, and measurement principles.

1. Introduction to Quantum Physics

What is Quantum Physics?

Quantum physics is the branch of physics that studies the behavior of matter and energy at the atomic and subatomic level (electrons, photons, atoms, etc.).

Unlike classical physics, particles in quantum physics behave differently. For example:

- An electron can behave like both a particle and a wave.
- A particle can exist in multiple states until it is measured.
- Two particles can remain connected even when they are far apart.

These unusual behaviors are used to build quantum computers.

Classical Computer vs Quantum Computer

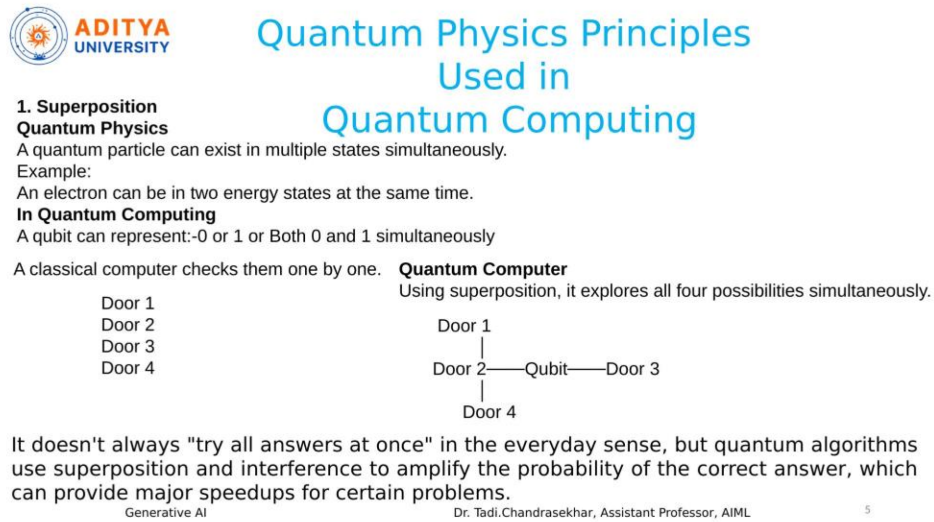
Classical Computer	**Quantum Computer**
Uses Bits	Uses Qubits
Bit = 0 or 1	Qubit = 0, 1, or both simultaneously
Works using classical electronics	Works using quantum physics
Uses electrical transistors	Uses atoms, ions, photons, or superconducting circuits

Quantum Physics Principles Used in Quantum Computing

1. Superposition

Quantum Physics: A quantum particle can exist in multiple states simultaneously. Example: an electron can be in two energy states at the same time.

In Quantum Computing: A qubit can represent 0, 1, or both 0 and 1 simultaneously. A classical computer checks possibilities one by one; using superposition, a quantum computer explores all possibilities simultaneously.



It doesn't always "try all answers at once" in the everyday sense, but quantum algorithms use superposition and interference to amplify the probability of the correct answer, which can provide major speedups for certain problems.

2. Entanglement

Quantum Physics: Two particles become linked. If one particle changes, the other changes instantly, even if they are very far apart.

In Quantum Computing: Qubits become entangled. This allows them to work together and perform complex calculations much faster.

3. Quantum Interference

Quantum Physics: Quantum waves can combine. Constructive interference increases the probability of correct answers; destructive interference reduces the probability of incorrect answers.

In Quantum Computing: Quantum algorithms use interference to make correct solutions more likely.

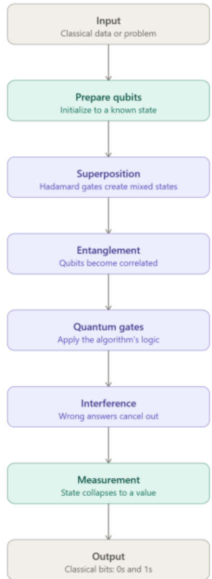
4. Wave-Particle Duality

Quantum particles behave as both a particle and a wave. This wave nature allows qubits to interfere with each other, which is essential for quantum computation.

5. Measurement Principle

Before measurement: A qubit exists in multiple possible states.

After measurement: The qubit collapses to a single state.



2. Historical Background

Historical Background of Quantum Mechanics

- Isaac Newton proposed that light is made of tiny particles called corpuscles.
- During the 19th century, experiments showed that light also behaves like a wave.
- Young's Double Slit Experiment proved the wave nature of light.
- Light was later identified as an electromagnetic wave.
- Classical wave theory could not explain blackbody radiation.
- In 1901, Max Planck introduced the idea that energy is emitted in small packets called quanta. Planck's idea marked the beginning of quantum mechanics. Planck proposed that atoms can have only discrete (fixed) energy levels.
- In 1905, Albert Einstein extended Planck's idea and proposed that light consists of photons. Each photon has energy given by $E = h\nu$, where h is Planck's constant. Einstein successfully explained the photoelectric effect using photons.
- In 1907, Einstein applied quantum theory to explain the specific heat of solids.
- In 1913, Niels Bohr proposed a model of the hydrogen atom with fixed energy levels.
- Bohr's model explained the hydrogen spectrum but had several limitations.
- Louis de Broglie proposed that matter also has wave properties. De Broglie's idea introduced the concept of wave-particle duality.
- Werner Heisenberg developed matrix mechanics to describe atomic systems.
- Erwin Schrödinger developed the wave equation for quantum mechanics.
- Max Born and Pascual Jordan improved Heisenberg's matrix formulation.
- Schrödinger's wave mechanics and Heisenberg's matrix mechanics together formed modern quantum mechanics.
- Around 1925, the Compton Effect provided strong evidence for the existence of photons.

Why These Topics Are Studied

Blackbody radiation, the photoelectric effect, and Compton scattering are studied because they could not be explained by classical physics.

These experiments led to the development of quantum mechanics by introducing the concepts of energy quantization, photons, and photon momentum.

They form the foundation of modern quantum physics and are essential for understanding advanced technologies such as lasers, solar cells, semiconductors, medical imaging, and quantum computing.

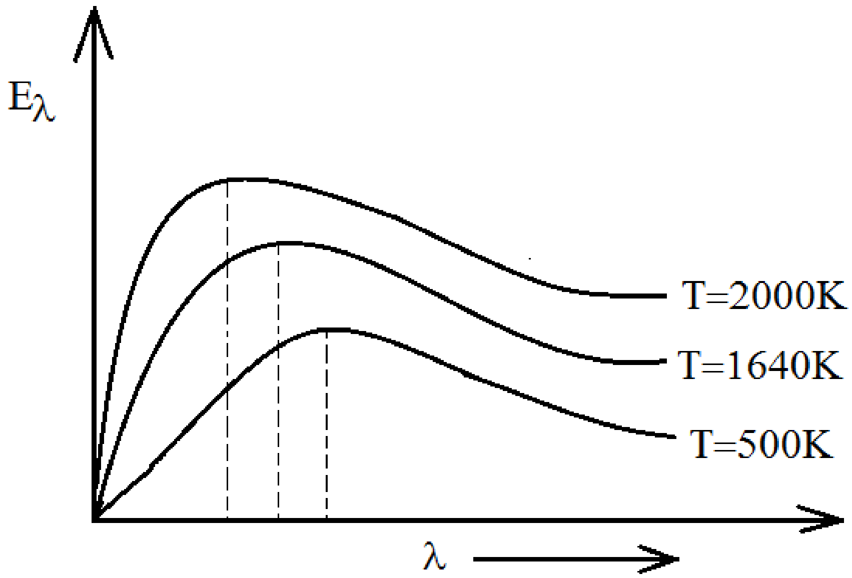
3. Blackbody Radiation

Blackbody radiation is the electromagnetic radiation emitted by a perfect black body, which absorbs all the light falling on it and emits radiation only based on its temperature.

As the temperature increases, the black body emits more energy and the peak of the emitted radiation shifts toward shorter wavelengths.

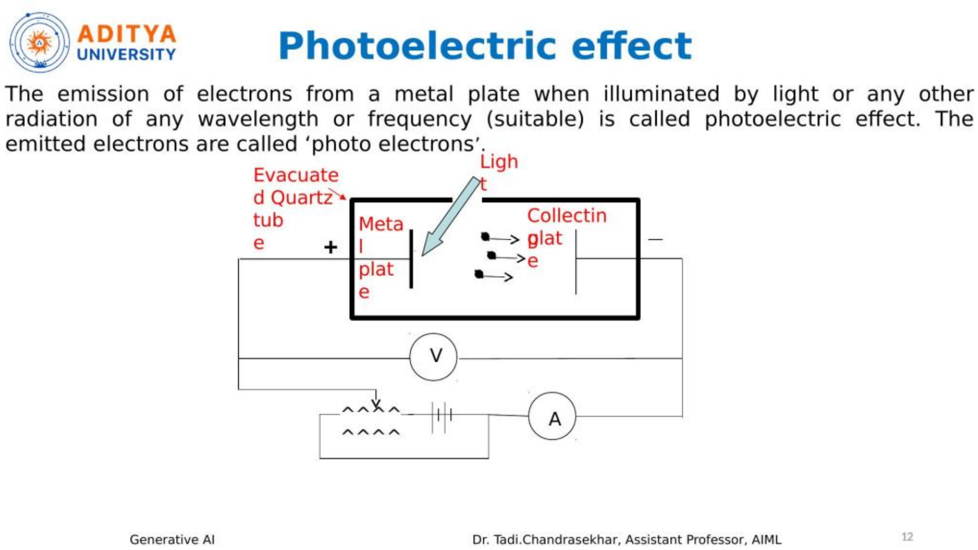
Classical theories such as the Stefan–Boltzmann law, Wien's law, and the Rayleigh–Jeans law could explain only parts of the blackbody radiation spectrum but failed to explain the complete energy distribution.

In 1901, Max Planck solved this problem by proposing that energy is emitted in discrete packets called quanta, with each quantum having energy $E = h\nu$, where h is Planck's constant. This idea of energy quantization successfully explained blackbody radiation and marked the beginning of quantum mechanics.



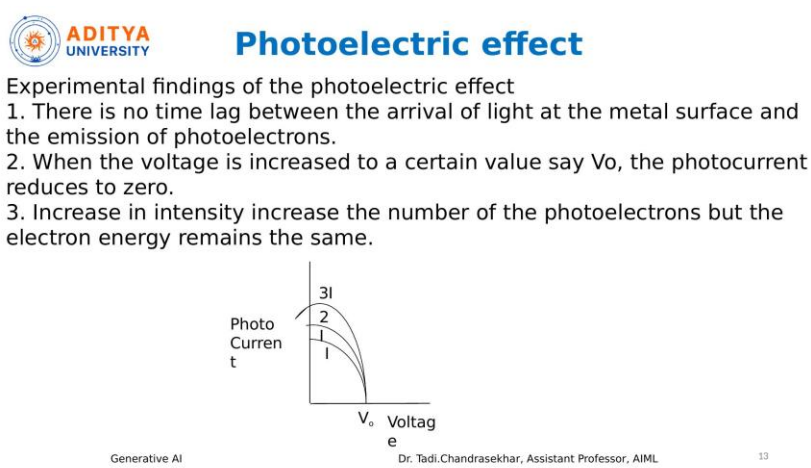
4. Photoelectric Effect

The emission of electrons from a metal plate when illuminated by light or any other radiation of suitable wavelength or frequency is called the photoelectric effect. The emitted electrons are called "photoelectrons."



Experimental Findings of the Photoelectric Effect

- There is no time lag between the arrival of light at the metal surface and the emission of photoelectrons.
- When the voltage is increased to a certain value, say V_0, the photocurrent reduces to zero.
- An increase in intensity increases the number of photoelectrons, but the electron energy remains the same.



Einstein's Photoelectric Explanation

The energy of an incident photon is utilized in two ways:

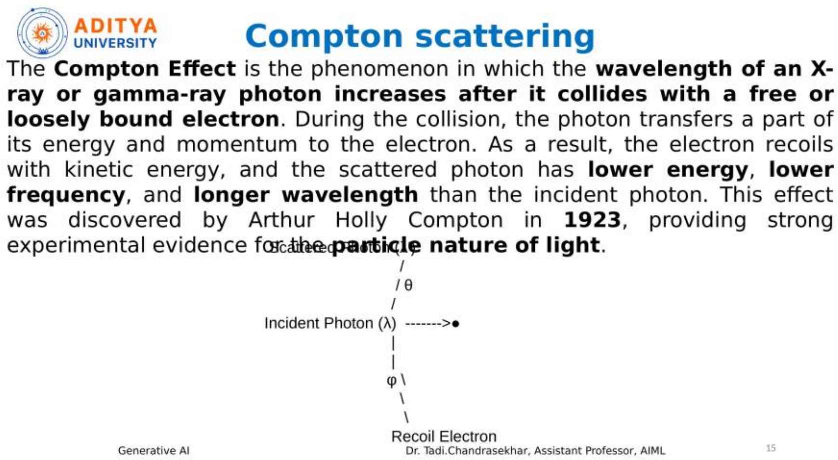
1. A part of the energy is used to free the electron from the atom, known as the photoelectric work function (W_0).
2. The other part is used in providing kinetic energy to the emitted electron ($\frac{1}{2}mv^2max$).

$$h\nu = W_0 + \frac{1}{2}mv^2max$$

This is called Einstein's photoelectric equation.

5. Compton Scattering

The Compton Effect is the phenomenon in which the wavelength of an X-ray or gamma-ray photon increases after it collides with a free or loosely bound electron. During the collision, the photon transfers a part of its energy and momentum to the electron. As a result, the electron recoils with kinetic energy, and the scattered photon has lower energy, lower frequency, and longer wavelength than the incident photon. This effect was discovered by Arthur Holly Compton in 1923, providing strong experimental evidence for the particle nature of light.



Compton Shift Derivation

Working Principle

1. A beam of **high-energy X-rays** is directed onto a target containing **free or loosely bound electrons** (such as graphite).
2. Each incident photon possesses energy

$$E = h\nu = \frac{hc}{\lambda}$$

where:

- h = Planck's constant
- ν = Frequency of the photon
- c = Speed of light
- λ = Wavelength of the incident photon

3. The photon collides with an electron that is initially at rest.
4. During the collision:
   - The photon transfers part of its **energy** and **momentum** to the electron.
   - The electron gains kinetic energy and moves away as a **recoil electron**.
   - The photon is scattered in another direction.
5. Since the photon loses some of its energy:
   - Its **frequency decreases**.
   - Its **wavelength increases**.

Hence,

$$\lambda' > \lambda$$

Compton Shift Equation

The increase in wavelength is called the **Compton Shift**.

$$\boxed{\Delta\lambda = \lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta)}$$

where

- $\Delta\lambda$ = Change in wavelength
- h = Planck's constant
- m_e = Mass of the electron
- c = Speed of light
- θ = Scattering angle of the photon

The quantity

$$\boxed{\frac{h}{m_e c} = 2.43 \times 10^{-12}\ \mathrm{m}}$$

is called the **Compton wavelength of the electron**.

Significance and Applications

The Compton Effect demonstrates that light possesses particle nature. When a photon collides with an electron, it transfers part of its energy and momentum to the electron, causing the photon to scatter with a longer wavelength and the electron to recoil. The experimentally verified Compton shift equation strongly supports the quantum theory of light and is one of the key experiments in the development of modern quantum mechanics.

Applications:

- X-ray scattering experiments.
- Medical imaging and radiography.
- Nuclear and particle physics.
- Astrophysics.
- Material characterization using X-rays.

6. Dual Nature of Light and Matter

The concept of wave-particle duality is one of the foundations of Quantum Mechanics. Classical physics considered light only as a wave and matter only as particles. However, experiments showed that both light and matter exhibit dual nature.

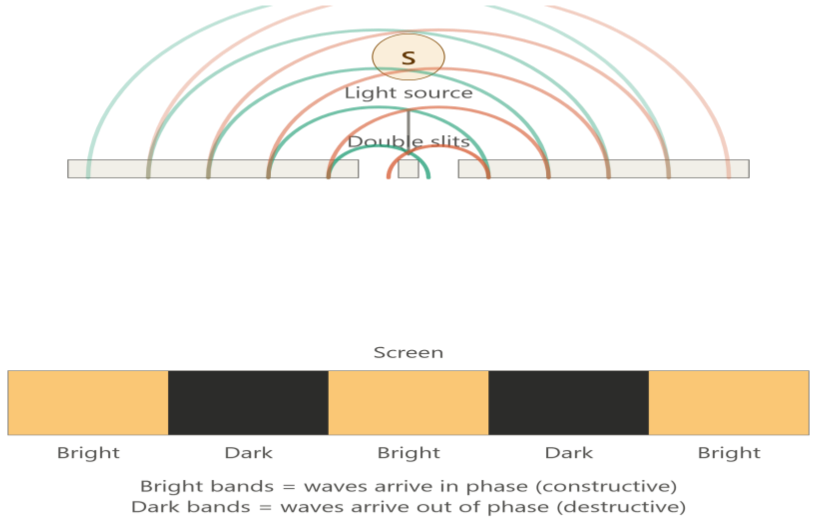
1. Dual Nature of Light

Light behaves as both a wave and a particle depending on the experiment performed.

(A) Light as a Wave

The following phenomena prove that light behaves like a wave:

- Interference
- Diffraction
- Polarization



Explanation: Light passes through two narrow slits. The waves overlap and produce bright and dark fringes. This proves that light behaves as a wave.

(B) Particle Nature of Light

The Photoelectric Effect, explained by Albert Einstein, proves that light is made up of tiny packets of energy called photons.

Photon Energy: $E = h\nu$, where E = Energy of photon, h = Planck's constant, ν = Frequency.

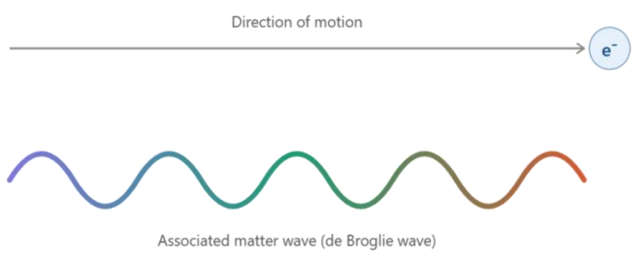
Explanation: When light falls on a metal surface, electrons are emitted. This shows that light transfers energy as particles (photons).

2. Dual Nature of Matter (de Broglie Hypothesis)

Scientists observed that light, which was once considered only a wave, also behaves like particles. Louis de Broglie (1924) proposed the reverse idea: if light has particle nature, then every moving particle should also possess wave nature. This idea is called the de Broglie Hypothesis.

Definition: Every moving material particle is associated with a wave called the Matter Wave or de Broglie Wave.

Examples: Electron, Proton, Neutron, Atom.



de Broglie Wavelength

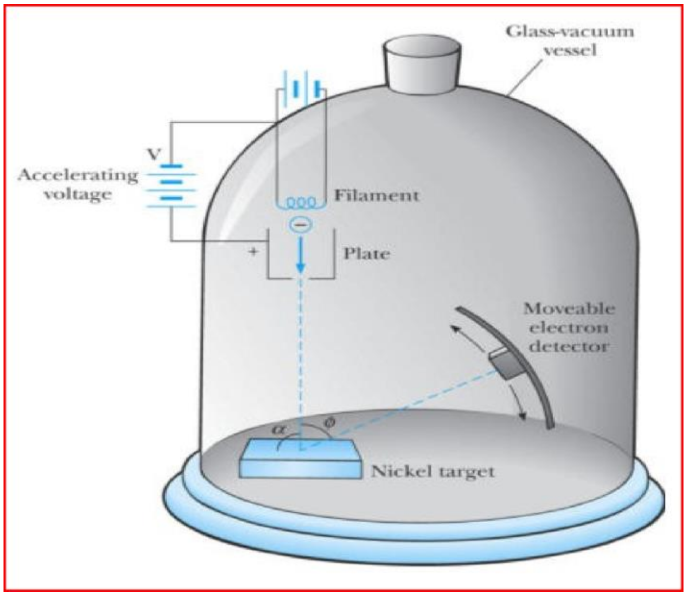
The wavelength associated with a moving particle is:

$$\lambda = h/p, \text{ and since } p = mv, \quad \lambda = h/mv$$

where λ = Wavelength, h = Planck's constant, m = Mass of particle, v = Velocity of particle.

Davisson–Germer Experiment

The Davisson–Germer experiment proved the wave nature of electrons.



Working:

- Electrons are accelerated towards a nickel crystal.
- The crystal acts like a diffraction grating.
- Electrons undergo diffraction.
- Diffraction is a property of waves. Hence, electrons also behave like waves.

7. Schrödinger Equation

This section derives Schrödinger's Time-Independent Wave Equation step by step, starting from de Broglie's matter-wave hypothesis.

Step 1: Matter Behaves Like a Wave

According to de Broglie's hypothesis, every moving particle (electron, proton, etc.) behaves like a wave. The wave associated with a moving particle is represented by the wave function:

Schrödinger's Time-Independent Wave Equation

1. Matter behaves like a wave

According to **de Broglie's hypothesis**, every moving particle (electron, proton, etc.) behaves like a wave.

The wave associated with a moving particle is represented by the **wave function**:

$$\boxed{\psi(x,t) = Ae^{-\frac{i}{\hbar}(Et-px)}}$$

where

- **ψ(x,t)** = Wave function (describes the particle)
- **A** = Maximum amplitude
- **E** = Energy of the particle
- **p** = Momentum
- **x** = Position
- **t** = Time
- **ħ** = Reduced Planck's constant

$$\hbar = \frac{h}{2\pi}$$

where ψ(x,t) = Wave function (describes the particle), A = Maximum amplitude, E = Energy of the particle, p = Momentum, x = Position, t = Time, ħ = Reduced Planck's constant (ħ = h/2π).

Step 2: Meaning of the Wave Function

The wave function ψ does not represent the particle itself. Instead, it contains all the information about the particle. The probability of finding the particle at a position is $|\psi|^2$.

2. Meaning of the Wave Function

The wave function ψ does **not** represent the particle itself.

Instead,

- It contains all the information about the particle.
- The probability of finding the particle at a position is

$$\boxed{|\psi|^2}$$

Thus,

- Large ψ → High probability
- Small ψ → Low probability

Thus: Large ψ → High probability; Small ψ → Low probability.

Step 3: Wave Function at Time t = 0

If we observe the particle at time t = 0, then Et = 0. Substituting into the wave equation, and using the de Broglie relation p = h/λ together with ħ = h/2π, gives p/ħ = 2π/λ.

Wave Function at $t = 0$

Final Equation

$$\boxed{\psi(x,0) = Ae^{\frac{2\pi i x}{\lambda}}}$$

Complete Form

$$\boxed{\psi(x,0) = Ae^{\frac{ipx}{\hbar}} = Ae^{\frac{2\pi i x}{\lambda}}}$$

Final (complete) form of the wave function at t = 0:

3. Wave Function at Time $t = 0$

If we observe the particle at **time = 0**, then

$$Et = 0$$

Substituting into the wave equation,

$$\psi(x,0) = Ae^{\frac{ipx}{\hbar}}$$

Using the de Broglie relation,

$$p = \frac{h}{\lambda}$$

Since

$$\hbar = \frac{h}{2\pi}$$

then

$$\frac{p}{\hbar} = \frac{h/\lambda}{h/2\pi} = \frac{2\pi}{\lambda}$$

Step 4: What Do the Symbols Mean?

Symbol	Meaning
ψ	Wave function

Symbol	Meaning
A	Maximum amplitude
λ	Wavelength
x	Position
t	Time
p	Momentum
E	Energy
h	Planck's constant
ħ	Reduced Planck's constant

Step 5: Differentiating the Wave Function

To derive Schrödinger's equation, we differentiate the wave function with respect to position x.

4. What do the symbols mean?

Symbol	Meaning
ψ	Wave function
A	Maximum amplitude
λ	Wavelength
x	Position
t	Time
p	Momentum
E	Energy
h	Planck's constant
ħ	Reduced Planck's constant

Since $i^2 = -1$, the second derivative simplifies to:

5. Differentiating the Wave Function

To derive Schrödinger's equation, we differentiate the wave function with respect to **position x**.

The wave function is

$$\psi = Ae^{\frac{2\pi i x}{\lambda}}$$

First derivative

Differentiate with respect to x:

$$\frac{\partial \psi}{\partial x} = \frac{2\pi i}{\lambda}\psi$$

Second derivative

Differentiate once again:

$$\frac{\partial^2 \psi}{\partial x^2} = \left(\frac{2\pi i}{\lambda}\right)^2 \psi$$

Since

$$i^2 = -1$$

we get

$$\boxed{\frac{\partial^2 \psi}{\partial x^2} = -\frac{4\pi^2}{\lambda^2}\psi}$$

Bringing in the Total Energy

The total energy of the particle is E = K + V, where K is kinetic energy and V is potential energy. The kinetic energy of a particle is K = $p^2/2m$, where p = momentum and m = mass of the particle.

The total energy is

$$\boxed{E = K + V}$$

where

- E = Total energy
- K = Kinetic energy
- V = Potential energy

Substituting into the total energy equation gives E = $p^2/2m$ + V.

The kinetic energy of a particle is

$$\boxed{K = \frac{p^2}{2m}}$$

where

- p = Momentum
- m = Mass of the particle

Substituting into the total energy equation,

$$\boxed{E = \frac{p^2}{2m} + V}$$

Rearranging to Obtain the Schrödinger Equation

Moving the potential energy to the left side: E − V = $p^2/2m$. Multiplying both sides by 2m: 2m(E − V) = p^2. Using ħ = h/2π (so $h^2 = 4\pi^2\hbar^2$) and substituting into the differentiated wave-function relation gives, after multiplying both sides by $-\hbar^2$:

Move the potential energy to the left side:

$$\boxed{E - V = \frac{p^2}{2m}}$$

Multiplying both sides by $2m$,

$$\boxed{2m(E - V) = p^2}$$

Equivalently, starting from E = $p^2/2m$ + V and multiplying both sides by the wave function ψ, then substituting $p^2\psi = -\hbar^2(d^2\psi/dx^2)$, we obtain the Time-Independent Schrödinger Equation (TISE):

$$\boxed{\hbar = \frac{h}{2\pi}},$$

then

$$h = 2\pi\hbar$$

and

$$h^2 = 4\pi^2\hbar^2.$$

Substitute into the equation:

$$\frac{d^2\psi}{dx^2} = -\frac{p^2}{\hbar^2}\psi$$

Multiply both sides by $-\hbar^2$:

$$\boxed{-\hbar^2\frac{d^2\psi}{dx^2} = p^2\psi}$$

$$E = \frac{p^2}{2m} + V$$

Rearranging,

$$2m(E - V) = p^2$$

Multiply both sides by the wave function ψ:

$$2m(E - V)\psi = p^2\psi$$

Using

$$p^2\psi = -\hbar^2\frac{d^2\psi}{dx^2},$$

we obtain

$$\boxed{-\hbar^2\frac{d^2\psi}{dx^2} = 2m(E - V)\psi}$$

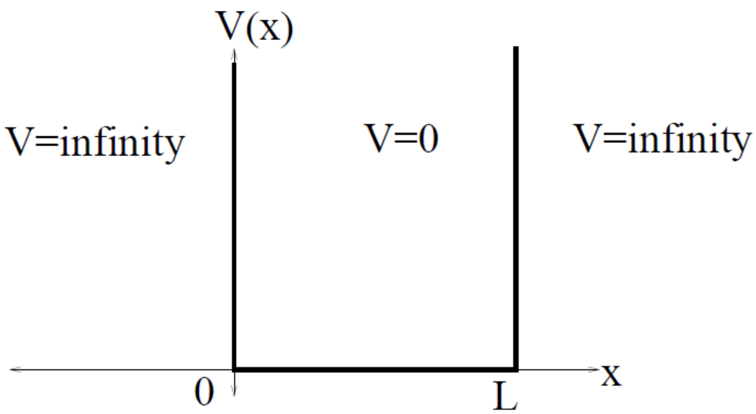
8. Quantum Mechanical Systems

Free Particle in an Infinite Potential Well

We confine the particle to a region between x = 0 and x = L. The potential (of infinite depth) is written as:

$$\begin{aligned} V(x) &= 0 \qquad \text{for} \quad 0 \le x \le L \\ &= \infty \qquad \text{otherwise} \end{aligned}$$

The potential energy is plotted as a function of a single variable, as shown below. The potential energy is 0 inside the box (V = 0 for 0 < x < L) and goes to infinity at the walls of the box (V = ∞ for x < 0 or x > L).



We assume the walls have infinite potential energy to ensure that the particle has zero probability of being at the walls or outside the box. This is necessary to apply the proper boundary conditions while solving the Time-Independent Schrödinger Equation (TISE) for an infinitely deep square well.

Solving the TISE for the Free Particle

The Time-Independent Schrödinger equation (TISE) for a particle of mass m moving in one direction with energy E is:

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2}(E - V)\psi = 0$$

This equation can be modified for a particle of mass m free to move parallel to the x-axis with zero potential energy (V = 0 everywhere), giving the standard wave-number form:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi$$

$$\frac{d^2\psi}{dx^2} + k^2\psi = 0$$

$$k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\psi(x) = A\sin(kx) + B\cos(kx)$$

This equation has the well-known solution $\psi(x) = A\sin(kx) + B\cos(kx)$. $\psi(x)$ determines the stationary states ($V = 0$) inside the box.

Applying Boundary Conditions

The boundary conditions require that the probability of finding the particle at $x = 0$ or $x = L$ is zero, which implies $\psi(x) = 0$ at those points.

When $x = 0$: $\sin(0) = 0$ and $\cos(0) = 1$, therefore:

$$\psi(0) = A\sin(0) + B\cos(0) = 0 \implies B = 0$$

Then for $x = L$, the following must be true:

$$\psi(L) = A\sin(kL) = 0 \implies kL = 0, \pi, 2\pi, 3\pi, \ldots, n\pi \quad (n = 1, 2, 3, \ldots)$$

Allowed Wave Functions

Allowed Wave Functions

For a particle confined in the region

$$0 \le x \le L,$$

the corresponding wave functions are

$$\psi_n(x) = A\sin\left(\frac{n\pi x}{L}\right), \qquad n = 1, 2, 3, \ldots$$

Outside the box, the probability of finding the particle is zero. Therefore,

$$\psi_n(x) = 0, \qquad x < 0 \quad \text{or} \quad x > L.$$

Determination of the Constant A

The solution still contains an unknown constant A. Although the wave function satisfies the Schrödinger equation for any value of A, only one value is physically acceptable. This value is obtained using the **normalization condition**, which ensures that the total probability of finding the particle inside the box is equal to **1**.

Normalization — Worked Calculation

The particle-in-a-box normalization integral requires evaluating:

$$\int_0^L |\psi_n(x)|^2\, dx = \int_0^L A^2 \sin^2(n\pi x/L)\, dx$$

Using a "dummy variable" $y = n\pi x/L$, so $x = (L/n\pi)y$ and $dx = (L/n\pi)dy$. The lower limit stays 0, but the upper limit becomes $y = (n\pi/L)\cdot L = n\pi$:

Let's carry out the calculations:

$$\int_0^L |\psi_n(x)|^2 dx = \int_0^L A^2 \sin^2\left(\frac{n\pi}{L}x\right)dx$$

Let's use a "dummy variable" $y \equiv \frac{n\pi}{L}x$

then $x = \frac{L}{n\pi}y$ and $dx = \frac{L}{n\pi}dy$

The lower limit is still 0, but the upper limit has to be changed to:

$$y_{\text{upper\_limit}} = \frac{n\pi}{L}x_{\text{upper\_limit}} = \frac{n\pi}{L}L = n\pi$$

$$= A^2 \frac{L}{n\pi}\int_0^{n\pi} \sin^2(y)dy$$

$$\boxed{\int_0^{n\pi} \sin^2(y)dy = \frac{n\pi}{2}}$$

So, we can continue from the preceding slide:

$$= A^2 \frac{L}{n\pi} \times \frac{n\pi}{2} = \frac{A^2 L}{2} = 1$$

From which we obtain immediately:

$$A = \sqrt{\frac{2}{L}}$$

and the complete solution for $\psi(x)$:

$$\psi(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi}{L}x\right)$$

Energy Eigenvalues

Energy of a Particle in an Infinite Potential Well (Simple Explanation)

Why do we calculate energy?

A particle trapped inside an infinite potential well cannot have any arbitrary energy. It can possess **only certain allowed energy values**, called **quantized energy levels**.

Relationship Between Wave Number and Energy

The energy of the particle is related to its wave number k by

$$k = \sqrt{\frac{2mE}{\hbar^2}}$$

or equivalently,

$$E = \frac{\hbar^2 k^2}{2m},$$

where

- m = mass of the particle
- $\hbar = \frac{h}{2\pi}$ = reduced Planck's constant
- k = wave number

Allowed Wave Numbers

From the boundary conditions of the box, only certain wave numbers are allowed:

$$k_n = \frac{n\pi}{L},$$

where

- L = width of the box
- $n = 1, 2, 3, \ldots$ = quantum number

Allowed Energy Levels

Substituting k_n into the energy equation,

$$E_n = \frac{\hbar^2}{2m}\left(\frac{n\pi}{L}\right)^2$$

which simplifies to

$$\boxed{E_n = \frac{n^2 h^2}{8mL^2}}, \qquad n = 1, 2, 3, \ldots$$

Step Potential in Quantum Mechanics

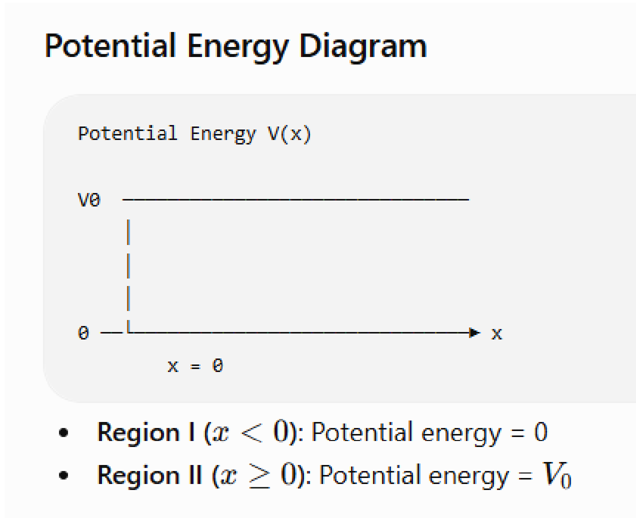
A **step potential** is a potential energy function in which the potential changes **suddenly** at a particular position, similar to climbing a step on a staircase.

Mathematically,

$$V(x) = \begin{cases} 0, & x < 0 \\ V_0, & x \geq 0 \end{cases}$$

where

- V_0 = Height of the potential step
- $x = 0$ = Position where the potential suddenly changes

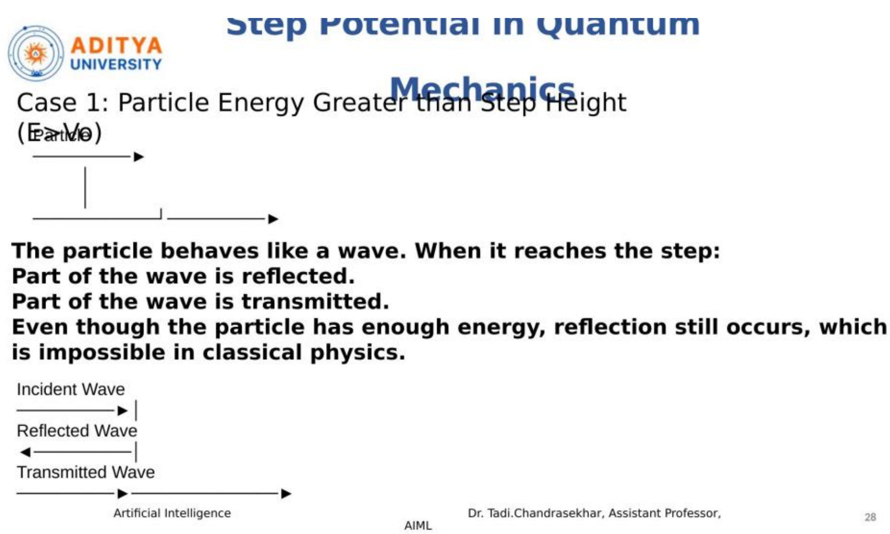


Case 1: Particle Energy Greater than Step Height ($E > V_0$)

The particle behaves like a wave. When it reaches the step:

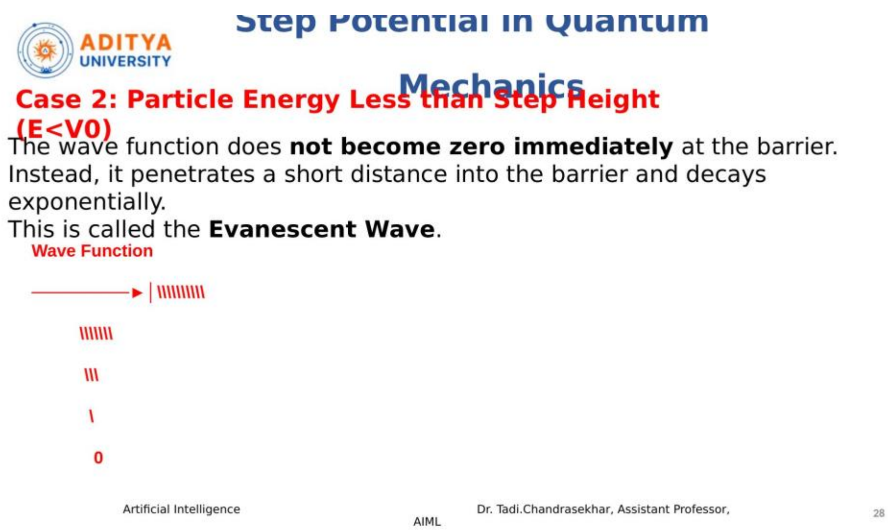
- Part of the wave is reflected.
- Part of the wave is transmitted.

Even though the particle has enough energy, reflection still occurs, which is impossible in classical physics.



Case 2: Particle Energy Less than Step Height ($E < V_0$)

The wave function does not become zero immediately at the barrier. Instead, it penetrates a short distance into the barrier and decays exponentially. This is called the Evanescent Wave.



9. Operators and Observables

In classical mechanics, physical quantities such as position, velocity, momentum, and energy are represented by ordinary numbers. However, in quantum mechanics, these quantities are represented by operators that act on the wave function.

1. What is an Operator?

Definition: An operator is a mathematical rule or function that acts on a wave function (ψ) to produce another function or to extract measurable physical information.

Mathematically,

$$\boxed{\hat{A}\psi}$$

where

- $\hat{A}$ = Operator
- ψ = Wave function

The symbol "^" **(hat)** indicates that A is an operator.

where Â = Operator, ψ = Wave function. The symbol "^" (hat) indicates that A is an operator.

2. What is an Observable?

An observable is any physical quantity that can be measured experimentally. Examples include: Position, Momentum, Energy, Angular momentum, and Spin. Each observable has a corresponding mathematical operator.

Observable	Operator
Position	$\hat{x}$
Momentum	$\hat{p}$
Energy	$\hat{H}$
Angular Momentum	$\hat{L}$

Example — Position Operator

The simplest operator is the position operator. It simply multiplies the wave function by the position x.

Mathematical Form

$$\boxed{\hat{x} = x}$$

If

$$\psi(x) = x^2$$

then

$$\hat{x}\psi = x(x^2) = x^3$$

Interpretation

The position operator gives information about the **position of the particle**.

Momentum Operator

Momentum is not represented by multiplication.

Instead, it is represented by a differential operator.

Momentum Operator

$$\boxed{\hat{p} = -i\hbar\frac{d}{dx}}$$

where

- $i = \sqrt{-1}$
- $\hbar = \dfrac{h}{2\pi}$

Hamiltonian Operator

The Hamiltonian operator represents the **total energy** of a quantum particle.

Classical Total Energy

$$E = \text{Kinetic Energy} + \text{Potential Energy}$$

or

$$E = \frac{p^2}{2m} + V(x)$$

Replacing momentum by its operator,

$$\boxed{\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)}$$

10. Commutation Relations and Uncertainty Principle

Commutator

What is a Commutator?

A **commutator** tells us whether two operators can be applied in any order.

For two operators $\hat{A}$ and $\hat{B}$, the commutator is defined as

$$\boxed{[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}}$$

where

- $\hat{A}$ = First operator
- $\hat{B}$ = Second operator
- $[\,,\,]$ = Commutator

Simple Meaning

A commutator compares two operations:

1. Apply operator **A first**, then **B**
2. Apply operator **B first**, then **A**

If both give the same result, the operators **commute**.

If they give different results, they **do not commute**.

Case 1: Operators Commute

If

$$\boxed{[\hat{A}, \hat{B}] = 0}$$

then

$$\hat{A}\hat{B} = \hat{B}\hat{A}$$

This means

- Order does not matter.
- Both observables can be measured simultaneously with perfect accuracy.

Case 2: Operators Do Not Commute

If

$$\boxed{[\hat{A}, \hat{B}] \neq 0}$$

then

$$\hat{A}\hat{B} \neq \hat{B}\hat{A}$$

This means

- Order matters.
- Simultaneous exact measurement is impossible.

Most Important Commutation Relation

The most important commutator in quantum mechanics is between the **position operator** and the **momentum operator**.

Position Operator

$$\boxed{\hat{x} = x}$$

Momentum Operator

$$\boxed{\hat{p} = -i\hbar \frac{d}{dx}}$$

Heisenberg Uncertainty Principle

Statement of Heisenberg Uncertainty Principle

Definition:

> **The product of the uncertainties in position and momentum of a particle can never be smaller than $\frac{\hbar}{2}$.**

Mathematically,

$$\boxed{\Delta x \, \Delta p \geq \frac{\hbar}{2}}$$

where

- Δx = Uncertainty in position (m)
- Δp = Uncertainty in momentum (kg·m/s)
- $\hbar = \frac{h}{2\pi}$ = Reduced Planck's constant

Meaning of the Equation

The equation

$$\Delta x\,\Delta p \geq \frac{\hbar}{2}$$

means that:

- If **position is measured very accurately** (Δx is very small), then the **uncertainty in momentum** (Δp) becomes very large.
- If **momentum is measured very accurately** (Δp is very small), then the **uncertainty in position** (Δx) becomes very large.

Thus, **both quantities cannot be known exactly at the same time.**

11. Quantum Postulates and Measurement

Postulate 1: State of a Quantum System

Every quantum system is completely described by a wave function (also called the state function). The wave function contains all the information about the quantum particle. It is represented by $\psi(x,t)$, or in Dirac notation $|\psi\rangle$, where x = Position, t = Time, and ψ = Wave function.

Postulate 2: Observables Are Represented by Operators

Every measurable physical quantity (called an observable) is represented by a mathematical operator. Examples of observables are Position, Momentum, Energy, Angular Momentum, and Spin.

Postulate 3: Measurement Gives Eigenvalues

When an observable is measured, the result is always one of the eigenvalues of the corresponding operator. Mathematically:

$$\hat{A}\psi = a\psi$$

where $\hat{A}$ = Operator, ψ = Eigenfunction, a = Eigenvalue.

Postulate 4: Wave Function Collapse

Before measurement, a quantum system may exist in a superposition of several possible states. After measurement, the wave function collapses into the eigenstate corresponding to the measured eigenvalue.

Postulate 5: Time Evolution of the Wave Function

The wave function changes with time according to the Schrödinger Equation. The Time-Dependent Schrödinger Equation is given by:

$$\boxed{i\hbar\frac{\partial\psi}{\partial t} = \hat{H}\psi}$$

where

- $\hat{H}$ = Hamiltonian operator
- ψ = Wave function

12. Summary

- Explained the historical foundations of Quantum Mechanics, including blackbody radiation, the photoelectric effect, Compton scattering, wave-particle duality, the de Broglie hypothesis, and the Schrödinger equation.
- Discussed quantum mechanical models, including the free particle, infinite potential well, step potential, operators, observables, commutation relations, and the Heisenberg uncertainty principle.
- Presented the quantum postulates, measurement theory, eigenvalues, and eigenfunctions for describing and analyzing quantum systems.

Suggested References

- David J. Griffiths, Introduction to Quantum Mechanics.
- Nouredine Zettili, Quantum Mechanics: Concepts and Applications.
- J.J. Sakurai, Modern Quantum Mechanics.
- Michael A. Nielsen and Isaac L. Chuang, Quantum Computation and Quantum Information.

Recommended Reading (from course syllabus):

- T1: "Quantum Computation and Quantum Information" by Michael A. Nielsen and Isaac L. Chuang.
- T2: "Quantum Mechanics: Concepts and Applications" by Nouredine Zettili.
- R1. Vedran Dunjko, Jacob M. Taylor and Hans J. Briegel, “Quantum-Enhanced Machine Learning”, Physical Review Letters 117 (13).
- R2. Maria Schuld and Nathan Killoran, Quantum Machine Learning in Feature Hilbert Spaces, Phys. Rev. Lett. 122.